

2. Frequency Domain Filtering

(a) Low-pass vs. high-pass filtering in the frequency domain

After the Fourier Transform, an image is represented as a combination of frequency components: low frequencies correspond to slowly varying regions (smooth areas, overall brightness) and high frequencies correspond to rapidly varying regions (edges, fine detail, noise).

Aspect	Low-pass filter	High-pass filter
Passes	Low frequencies	High frequencies
Blocks	High frequencies	Low frequencies
Effect	Smoothing / blurring	Sharpening / edge enhancement
Use case	Noise reduction	Edge / detail detection

(b) Low-pass filtering — retain only the first two components

No specific frequency values were supplied, so the sample set below (ordered from lowest to highest frequency) is used.

Component	F1	F2	F3	F4	F5	F6
Original	120	45	30	15	8	3
Filtered (low-pass)	120	45	0	0	0	0

Filtered output = [120, 45, 0, 0, 0, 0]. The two lowest-frequency components are preserved and all higher-frequency components are set to zero, which removes fine detail and noise while retaining the overall image structure.